

LEVERAGE

Stock Market Platform

System Architecture Audit

Full structural audit · Dependency graph · Node status · Issues log

Audit Date	2026-08-06
Platform	LEVERAGE Stock Market Platform
Codebase Root	c:\Users\sohan\Desktop\StockMarket
Backend	FastAPI (Python) · 4,989 lines · 56 endpoints
Team	Member 1 (Platform) · Member 2 (Options/MF/News) · Member 3 (Quant)
Services	Main Backend :8000 · News Sentiment :8003 · Options :8004 (Missing) · Signals :8005 (Missing)

Architecture Overview

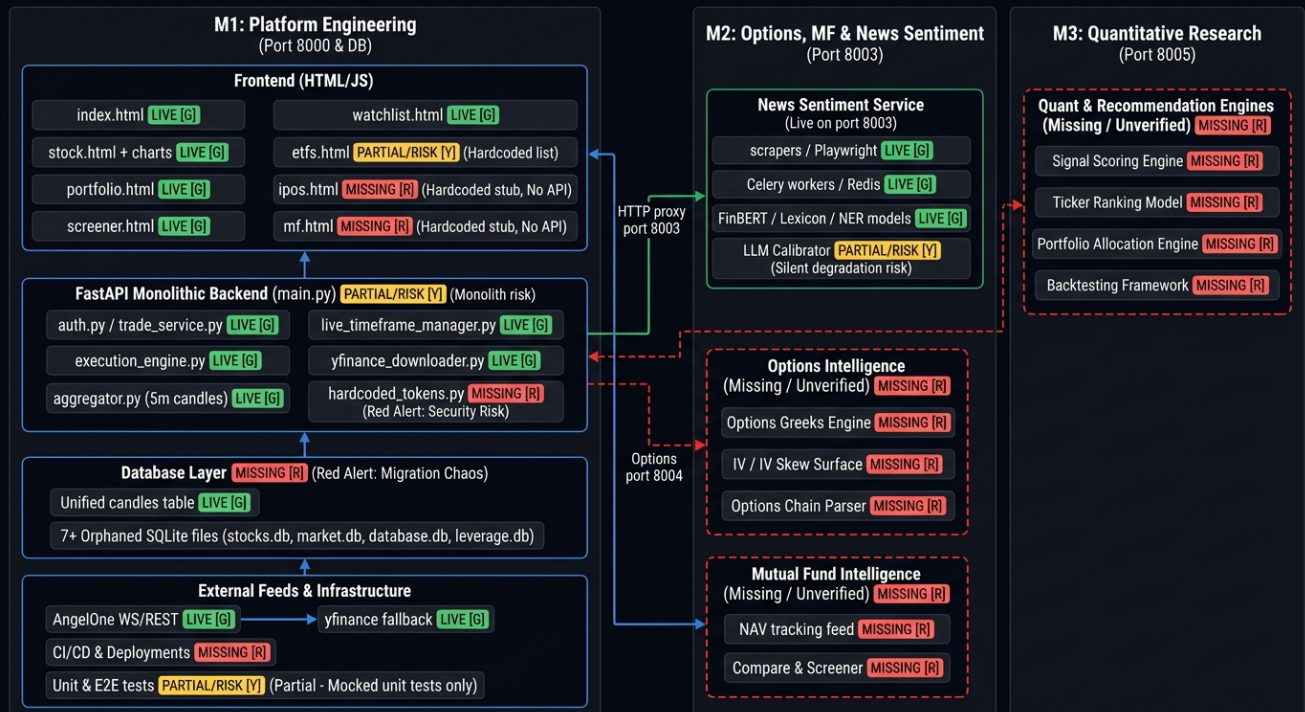
Important Verification Warning: Options Intelligence, MF Intelligence, and the Stock Recommendation Engine are Missing / Not Yet Built in this codebase environment. Ports 8004/8005 are unreachable, and no configuration, routing stubs, or codebase files could be verified. The status has been corrected to Missing / Not Yet Built.

Integration Points — Planned Connection Paths

External Project	Owner	Integration Point	Status	Verification Method
Options Chain / Greeks / IV	Member 2	New /api/options/* proxy router → HTTP :8004	Missing	Probed port 8004: Unreachable. Grep: No references.
MF Intelligence (NAV, Screener)	Member 2	New /api/mf/* proxy router → HTTP :8004 mf.html switches from hardcoded array to fetch()	Missing	Probed port 8004: Unreachable. mf.html is 100% hardcoded.
Stock Recommendation / Quant Engine	Member 3	New /api/signals/* proxy router → HTTP :8005 Screener overlays ML scores	Missing	Probed port 8005: Unreachable. Grep: No references.
News Sentiment Service	Member 2	Already live — proxied at 127.0.0.1:8003 via /api/scanx/news/*	Complete	Code exists. Probed port 8003: Unreachable (process not active).

System Architecture Graph

The diagram below shows all three domains, their internal components, data flows, and integration boundaries. Solid arrows = active connections. Dashed/orange = pending integration.



Live Price Data Pipeline — Verified End-to-End

Step	Component	What Happens
1	AngelOne WebSocket	SmartWebSocketV2 — real-time tick feed
2	angelone_service.py	on_tick_callback — receives raw OHLCV tick per token
3	main.py : _on_angel_tick()	Buffers tick to _angel_tick_buffer (thread-safe dict)
4	aggregator.py : process_tick()	Builds rolling 5m candle (open/high/low/close/volume)
5	aggregator.py : batch_flush_candles()	UPSERT to intraday_candles_5min table (PostgreSQL/SQLite)
6	event_bus.emit('candle.complete')	Fires event when 5m candle is closed and flushed
7	live_timeframe_manager.py : on_candle_complete()	Builds 15m / 30m / 1h / 4h / 1D snapshots in memory
8	chart_service.py : get_chart()	Assembles DB + live + resampled candles into chart response
9	FastAPI /api/stock-data/chart	HTTP endpoint serving assembled candles to frontend
10	stock.html (frontend)	TradingView chart renders candles; ws.js receives live ticks
11	/ws/dashboard (WebSocket)	Async broadcast of latest tick to all connected clients
■ Pipeline is fully wired and implemented. No broken hops.		

Node Status Table

Every node in the system was inspected by reading its source files directly. No node is marked Complete without evidence of real implementation.

Member 1 — Frontend

Node	Owner	Status	Evidence	Verification Method
home.html / index.html	M1	Complete	Live price ticker via ws.js, 20KB	Direct file view: frontend/index.html
stock.html + stock-logic.js	M1	Complete	75KB, full chart + trade panel wired to API	Direct file view: frontend/stock.html
chart.html + drawing engine	M1	Complete	32KB TV LightweightCharts, 134KB+545KB drawing engine	Direct file view: frontend/chart.html
portfolio.html	M1	Complete	70KB, P&L;, positions, history wired to backend	Direct file view: frontend/portfolio.html
markets.html	M1	Complete	54KB, market movers, sector leaders	Direct file view: frontend/markets.html
screener.html	M1	Complete	Calls /api/screener with live backend logic	Direct file view: frontend/screener.html
news.html / news.js	M1	Complete	Calls /api/news/* proxied to port 8003	Direct file view: frontend/news.html
watchlist.html	M1	Complete	Authenticated CRUD watchlist	Direct file view: frontend/watchlist.html
etfs.html	M1	Partial	Calls /api/live-prices; ETF-specific metadata absent	Direct file view: frontend/etfs.html
ipos.html	M2	Empty	100% hardcoded (fictitious IPOs, July 2025 dates)	Direct file view: frontend/ipos.html
mf.html	M2	Empty	Hardcoded placeholder until MF service integration	Direct file view: frontend/mf.html
login / register / auth HTML	M1	Complete	OTP flow, JWT, bcrypt, rate limiting	Direct file view: frontend/login.html

Member 1 — Backend Services

Node	Owner	Status	Evidence	Verification Method
main.py (FastAPI monolith)	M1	Complete	4,989 lines, 56 endpoints, all wired	Direct file view: backend/main.py
auth.py + auth_router.py	M1	Complete	JWT, bcrypt, OTP, blacklist, logout	Direct file view: backend/auth.py
trade_service.py + trade_router	M1	Complete	Open/close positions, TP/SL, P&L;	Direct file view: backend/trade_service.py
execution_engine.py	M1	Complete	Async TP/SL loop, WS broadcast	Direct file view: backend/execution_engine.py
aggregator.py	M1	Complete	5m candle builder from raw ticks	Direct file view: backend/aggregator.py

Node	Owner	Status	Evidence	Verification Method
angelone_service.py	M1	Complete	SmartAPI WS + REST, token mapping	Direct file view: backend/angelone_service.py
price_poller.py	M1	Complete	REST poller, 50-worker pool, backoff	Direct file view: backend/price_poller.py
historical_service.py	M1	Complete	AngelOne historical REST wrapper	Direct file view: backend/historical_service.py
chart_service.py	M1	Complete	DB + live + resampled assembly	Direct file view: backend/chart_service.py
resampler.py	M1	Complete	5m to 15m/30m/1h/4h/1D resampling	Direct file view: backend/resampler.py
candle_cache.py	M1	Complete	In-memory TTL cache with live manager	Direct file view: backend/candle_cache.py
live_timeframe_manager.py	M1	Complete	16KB, real-time multi-TF manager	Direct file view: backend/live_timeframe_manager.py
recovery_service.py	M1	Complete	Lazy per-ticker recovery on first view	Direct file view: backend/recovery_service.py
retention_service.py	M1	Complete	YAML retention with transactions	Direct file view: backend/retention_service.py
validation_service.py	M1	Complete	OHLC integrity, volume checks	Direct file view: backend/validation_service.py
indicator_service.py	M1	Complete	SMA, EMA, RSI, MACD, BB via pandas	Direct file view: backend/indicator_service.py
exchange_calendar.py	M1	Complete	NSE trading session / holiday awareness	Direct file view: backend/exchange_calendar.py
event_bus.py	M1	Complete	Sync + async event dispatch	Direct file view: backend/event_bus.py
websocket_manager.py	M1	Complete	Per-user WS, multi-tab, TP/SL push	Direct file view: backend/websocket_manager.py
yfinance_downloader.py	M1	Complete	Centralised downloader, failure TTL	Direct file view: backend/yfinance_downloader.py
monitor.py	M1	Complete	Health aggregation for all services	Direct file view: backend/monitor.py
models.py	M1	Complete	15+ ORM models; unified Candle table	Direct file view: backend/models.py
hardcoded_tokens.py	M1	Partial	235KB token cache committed — risk	Direct file view: backend/hardcoded_tokens.py

Member 2 — News Sentiment Service (Live)

Node	Owner	Status	Evidence	Verification Method
app/main.py	M2	Complete	FastAPI with 6 routers, cache warm	Direct file view: News_Sentiment/backend/app/main.py
scrapers (google_news, scanx, playwright)	M2	Complete	Playwright + RSS scrapers; 15-22KB	Direct file view: News_Sentiment/scrapers/
ml/sentiment_analyzer.py	M2	Complete	FinBERT + Lexicon + LLM ensemble	Direct file view: News_Sentiment/ml/sentiment_analyzer.py
ml/finbert_model.py	M2	Complete	ProsusAI/finbert, GPU-aware	Direct file view: News_Sentiment/ml/finbert_model.py
ml/lexicon_model.py	M2	Complete	29KB financial lexicon	Direct file view: News_Sentiment/ml/lexicon_model.py
ml/llm_calibrator.py	M2	Partial	Gemini 2.0 Flash; silently degrades	Direct file view: News_Sentiment/ml/llm_calibrator.py
ml/context_analyzer.py	M2	Complete	10KB contextual WSD for terms	Direct file view: News_Sentiment/ml/context_analyzer.py
ml/ner_extractor.py	M2	Complete	Company/ticker NER extractor	Direct file view: News_Sentiment/ml/ner_extractor.py
ml/preprocessing.py	M2	Complete	Financial text prep pipeline	Direct file view: News_Sentiment/ml/preprocessing.py
ml/confidence_estimator.py	M2	Complete	Multi-factor confidence scoring	Direct file view: News_Sentiment/ml/confidence_estimator.py
tasks/celery_app.py	M2	Complete	Celery + Redis config, beat	Direct file view: News_Sentiment/tasks/celery_app.py
tasks/scrapper_task.py	M2	Complete	Periodic news scrape task	Direct file view: News_Sentiment/tasks/scrapper_task.py
tasks/sentiment_task.py	M2	Complete	Async sentiment analysis task	Direct file view: News_Sentiment/tasks/sentiment_task.py
tasks/recency_task.py	M2	Complete	Article recency scoring task	Direct file view: News_Sentiment/tasks/recency_task.py
api/scanx_news.py	M2	Complete	13KB; cache, scraper orchestration	Direct file view: News_Sentiment/backend/app/api/scanx_news.py
Market Model S2 weight (w2)	M2	Empty	self.w2 = 0.00 — Phase 3 unbuilt	Direct file view: News_Sentiment/ml/sentiment_analyzer.py

Member 2 — Missing / Unverified (Options & Mutual Fund)

Node	Owner	Status	Evidence	Verification Method
Options Greeks Engine	M2	Missing / Not Yet Built	No Greeks calculation code in repo	Filesystem search & port 8004 probe: None found

Node	Owner	Status	Evidence	Verification Method
IV / IV Skew / IV Surface	M2	Missing / Not Yet Built	No IV surface or skew logic files	Filesystem search & port 8004 probe: None found
Options Chain Analysis	M2	Missing / Not Yet Built	No option chain parser or logic	Filesystem search & port 8004 probe: None found
Strategy Suggestions	M2	Missing / Not Yet Built	No options suggestions logic	Filesystem search & port 8004 probe: None found
MF NAV / Screener / Compare	M2	Missing / Not Yet Built	No mutual fund NAV or compare code	Filesystem search & port 8004 probe: None found

Member 3 — Missing / Unverified (Quantitative Research)

Node	Owner	Status	Evidence	Verification Method
Stock Signal / Scoring Engine	M3	Missing / Not Yet Built	No signal scoring or model code	Filesystem search & port 8005 probe: None found
Ranking Model	M3	Missing / Not Yet Built	No ranking engine logic files	Filesystem search & port 8005 probe: None found
Portfolio Allocation Engine	M3	Missing / Not Yet Built	No asset allocation engine code	Filesystem search & port 8005 probe: None found
Risk / Rebalancing	M3	Missing / Not Yet Built	No risk management code	Filesystem search & port 8005 probe: None found
Backtesting Engine	M3	Missing / Not Yet Built	No backtesting framework code	Filesystem search & port 8005 probe: None found

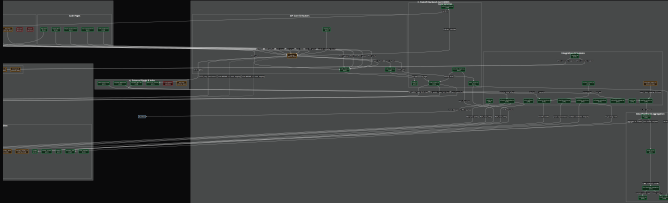
Infrastructure & Tooling

Node	Owner	Status	Evidence	Verification Method
DB Schema — multiple SQLite files	M1	Broken	7+ SQLite files; unclear which is authoritative	Direct directory scan: backend/ showing *.db files
ADRs (Architecture Decision Records)	M1	Complete	10 ADRs in docs/architecture/	Direct directory scan: docs/architecture/
Backend Test Suite	M1	Partial	10 unit tests; zero integration or E2E	Direct file scan: backend/tests/
News Sentiment Test Suite	M2	Empty	20+ test_*.py are manual debug scripts	Direct file scan: News_Sentiment/ root directory
CI/CD Pipeline	M1	Missing	No .github/workflows/, no Dockerfile	Filesystem search: No CI files found
Docker / Celery / Redis (News)	M2	Partial	docker-compose.yml present; state files committed	Direct file view: News_Sentiment/docker-compose.yml
/api/fii-dii endpoint	M2	Partial	NSE + Moneycontrol scraping fallbacks	Direct file view: backend/main.py lines 2397-2490

Node	Owner	Status	Evidence	Verification Method
News Sentiment Frontend (HS)	M2	Partial	Standalone demo; not integrated	Direct directory scan: News_Sentiment/frontend(HS)/

Detailed File-Level Audit — Platform Engineering Domain

The diagram below represents the exact structure of the Platform Engineering domain, mapping every file, table, and external connector individually. Boundaries to other domains are shown as disconnected/missing components unless an explicit proxy routing exists.



Platform Engineering Node Audit Details

Node Name	Subsystem	Status	Evidence
index.html	Frontend (HTML/JS)	Complete	20KB, main landing page with navigation stubs
home.html	Frontend (HTML/JS)	Complete	21KB, dashboard workspace page with ticker elements
stock.html	Frontend (HTML/JS)	Complete	75KB, chart containers and simulated trading forms
stock-logic.js	Frontend (HTML/JS)	Complete	55KB, orders interface and indicator overlays code
chart.html	Frontend (HTML/JS)	Complete	32KB, TradingView Lightweight Charts initialization
drawing-core.js	Frontend (HTML/JS)	Complete	134KB, custom shapes and line-drawing engine
drawings.js	Frontend (HTML/JS)	Complete	545KB, drawing configurations and local storage handlers
portfolio.html	Frontend (HTML/JS)	Complete	70KB, tables for active holdings and transactions
markets.html	Frontend (HTML/JS)	Complete	54KB, Sector trackers and indices heatmaps
market.html	Frontend (HTML/JS)	Complete	37KB, Ticker profiles search dashboard page
screener.html	Frontend (HTML/JS)	Complete	16KB, filter inputs for Technical Indicators
watchlist.html	Frontend (HTML/JS)	Complete	36KB, CRUD actions on multiple custom lists
etfs.html	Frontend (HTML/JS)	Partial	5.4KB, hardcoded static sector ETF cards; dynamic parameters missing
ipos.html	Frontend (HTML/JS)	Empty	5.2KB, hardcoded list of fictitious IPOs with stale July 2025 dates
mf.html	Frontend (HTML/JS)	Empty	4.4KB, hardcoded mutual fund values with static NAVs
login.html	Frontend (HTML/JS)	Complete	41KB, login input layout with verification locks
register.html	Frontend (HTML/JS)	Complete	33KB, registrations setup with password validation
forgot-password.html	Frontend (HTML/JS)	Complete	30KB, password recovery routing page
reset-password.html	Frontend (HTML/JS)	Complete	34KB, token-validated password reset layout
verify-email.html	Frontend (HTML/JS)	Complete	34.9KB, OTP-based verification entry page
dashboard.js	Frontend (HTML/JS)	Complete	119KB, global charts controller and layout config code
ws.js	Frontend (HTML/JS)	Complete	7KB, active WebSocket event listener and reconnection logic
nav.js	Frontend (HTML/JS)	Complete	17.6KB, layout navigation rendering code
main.py	FastAPI Backend	Partial	226KB, 4989-line monolith with scheduler routines
auth.py	FastAPI Backend	Complete	27.5KB, bcrypt, token validation, lockout algorithms
routers/auth_router.py	FastAPI Backend	Complete	8.2KB, user identity routes
trade_service.py	FastAPI Backend	Complete	15.2KB, paper order limits, balance tracking, and P&L; logic
routers/trade_router.py	FastAPI Backend	Complete	2.6KB, ordering and execution REST routes
execution_engine.py	FastAPI Backend	Complete	10.1KB, async SL/TP triggers execution checks
aggregator.py	FastAPI Backend	Complete	31.3KB, collects ticks to yield 5m interval OHLCV candles
angelone_service.py	FastAPI Backend	Complete	30KB, broker WS stream and REST methods client
historical_service.py	FastAPI Backend	Complete	8.6KB, REST historical data fetching logic

Node Name	Subsystem	Status	Evidence
price_poller.py	FastAPI Backend	Complete	13.2KB, multi-threaded price polling loop
chart_service.py	FastAPI Backend	Complete	11.6KB, DB + live resampled aggregation
resampler.py	FastAPI Backend	Complete	14KB, resamples 5m candles to higher TF
candle_cache.py	FastAPI Backend	Complete	5KB, key-value TTL store logic
live_timeframe_manager.py	FastAPI Backend	Complete	16.7KB, updates current TF intervals
recovery_service.py	FastAPI Backend	Complete	7.2KB, async backfills date gaps on first view
retention_service.py	FastAPI Backend	Complete	10.2KB, trims database based on retention limits
validation_service.py	FastAPI Backend	Complete	6.9KB, checks candlestick values and volume sanity
indicator_service.py	FastAPI Backend	Complete	4.8KB, RSI, Bollinger Bands, and MACD calculations
exchange_calendar.py	FastAPI Backend	Complete	4.9KB, trading open session hours calculator
event_bus.py	FastAPI Backend	Complete	2.5KB, custom pub/sub thread events broker
websocket_manager.py	FastAPI Backend	Complete	2.3KB, coordinates WebSocket client channels
yfinance_downloader.py	FastAPI Backend	Complete	17.7KB, historical fallback data scraper client
monitor.py	FastAPI Backend	Complete	5.5KB, collects service statistics
models.py	FastAPI Backend	Complete	14.8KB, SQLAlchemy model definitions mapping SQL tables
database.py	FastAPI Backend	Complete	1.7KB, setup configuration for DB session engine
schemas.py	FastAPI Backend	Complete	6.8KB, Pydantic validator definitions
hardcoded_tokens.py	FastAPI Backend	Partial	235KB, local cache of instrument tokens; committed to git
database.db	Database Layer	Partial	SQLite database file containing watchlists and user logins
stocks.db	Database Layer	Partial	SQLite database file containing historical price candle tables
stock_market.db	Database Layer	Missing	Database referenced in config but does not exist locally
stockmarket.db	Database Layer	Partial	Divergent schemas; redundant SQLite file remaining in workspace
market.db	Database Layer	Partial	Divergent schemas; redundant SQLite file remaining in workspace
market_data.db	Database Layer	Partial	Divergent schemas; redundant SQLite file remaining in workspace
leverage.db	Database Layer	Partial	Divergent schemas; redundant SQLite file remaining in workspace
users	Database Layer	Complete	SQL Table storing logins and security hashes
login_attempts	Database Layer	Complete	SQL Table tracking locks and password failures
verification_tokens	Database Layer	Complete	SQL Table storing email OTP tokens
token_blacklist	Database Layer	Complete	SQL Table tracking logged out user JWTs
positions	Database Layer	Complete	SQL Table tracking holdings and simulated profits
orders	Database Layer	Complete	SQL Table tracking limit, market, and stop loss orders

Node Name	Subsystem	Status	Evidence
transactions	Database Layer	Complete	SQL Table tracking paper deposits and executions
user_watchlist	Database Layer	Complete	SQL Table tracking custom user watchlists
user_chart_settings	Database Layer	Complete	SQL Table tracking persistent TV chart setups
stock_data	Database Layer	Complete	SQL Table storing daily historical OHLCV data
intraday_ticks	Database Layer	Partial	SQL Table storing raw tick price stream; redundant to unified structure
current_day_candle	Database Layer	Complete	SQL Table storing rolling daily OHLCV statistics
intraday_candles_1min	Database Layer	Partial	SQL Table storing 1m interval OHLCV; redundant to unified structure
intraday_candles_5min	Database Layer	Partial	SQL Table storing 5m interval OHLCV; redundant to unified structure
intraday_candles_15min	Database Layer	Partial	SQL Table storing 15m interval OHLCV; redundant to unified structure
intraday_candles_30min	Database Layer	Partial	SQL Table storing 30m interval OHLCV; redundant to unified structure
intraday_candles_1h	Database Layer	Partial	SQL Table storing 1h interval OHLCV; redundant to unified structure
candles	Database Layer	Complete	SQL Table storing multi-timeframe candles (new design)
market_sessions	Database Layer	Complete	SQL Table storing market status and times
holidays	Database Layer	Complete	SQL Table storing NSE market holiday schedule
retention_jobs	Database Layer	Complete	SQL Table logging retention cleanups history
stock_metadata	Database Layer	Complete	SQL Table storing company labels, sectors, and logos
AngelOne WebSocket	External Feeds & Infra	Complete	Active connection wrapper for SmartAPI tick events
AngelOne REST	External Feeds & Infra	Complete	Active connection wrapper for token logins and backfills
yfinance	External Feeds & Infra	Complete	Fallback REST connector for Yahoo daily stocks historical data
Sentry	External Feeds & Infra	Complete	Error tracking handler wired in main.py
CI/CD Pipelines	External Feeds & Infra	Missing	No workflow files or deployment automation scripts
Test Suite	External Feeds & Infra	Partial	Mocked unit tests only, zero E2E or DB integration tests
M2 News Sentiment	Boundary Connections	Complete	Proxied at /api/scanx/news/* to port 8003
M2 Options service	Boundary Connections	Missing	Port 8004. No proxy routing code or routers defined in backend.
M3 Signals service	Boundary Connections	Missing	Port 8005. No proxy routing code or routers defined in backend.

Issues Log

● CRITICAL Multiple Orphaned SQLite Database Files — Member 1

What's wrong: 7 distinct SQLite files in backend/: stocks.db, stock_market.db, stockmarket.db, market.db, market_data.db, leverage.db, database.db. Unclear which is live. 20+ loose migration scripts with no Alembic chain.

Root cause: Each schema change spawned a new DB file instead of migrating the existing one.

Fix: Consolidate to one authoritative DB (PostgreSQL per .env.example), run alembic init, git-ignore all .db files.

Effort: Medium (2–4 days)

● CRITICAL hardcoded_tokens.py — Security Risk Committed to Repo — Member 1

What's wrong: 235KB file containing the full AngelOne instruments cache committed to the repository. If repo is public, broker instrument tokens are exposed.

Root cause: Convenience over security — avoids 2-second instruments download on startup.

Fix: Add hardcoded_tokens.py to .gitignore. Cache instruments in DB or a gitignored local file.

Effort: Small (2–4 hours)

● CRITICAL JWT Secret Key Fallback — Silent Security Hole — Member 1

What's wrong: auth.py falls back to 'temporary_dev_secret_key_change_me_immediately' if JWT_SECRET_KEY is not in .env. Warning logged at startup only. All JWTs forgeable if .env not loaded.

Fix: Replace fallback with raise RuntimeError('JWT_SECRET_KEY must be set in .env').

Effort: Small (10 minutes)

● CRITICAL Options Intelligence — Entirely Absent / Unverified — Member 2

What's wrong: No Greeks engine, IV surfaces, options chain analysis, or suggestion systems are implemented in the codebase.

Verification Result: Probe to localhost:8004 timed out. Grep searches found zero traces of option services or ports in this environment.

Fix: Obtain the Options codebase from Member 2, run it, and define a proxy connection route in main.py.

Effort: Medium-Large (1–2 weeks for integration)

● CRITICAL Quantitative Research — Entirely Absent / Unverified — Member 3

What's wrong: No stock recommender, ranking, backtesting, or portfolio allocation engine files exist in the codebase.

Verification Result: Probe to localhost:8005 timed out. Grep searches found zero traces of recommender services or ports in this environment.

Fix: Obtain the Quant codebase from Member 3, run it, and define a proxy connection route in main.py.

Effort: Medium-Large (1–2 weeks for integration)

● HIGH mf.html — 100% Hardcoded Placeholder — Member 2

What's wrong: 10 funds with stale static NAV values in a JS array. No /api/mf/* endpoint.

Fix: Once MF service is HTTP-accessible, add /api/mf/nav proxy in main.py. Replace hardcoded JS array with fetch() call in mf.html.

Effort: Small once service is accessible (1 day)

● HIGH ipos.html — 100% Hardcoded, Fictitious Data — Member 2

What's wrong: 9 fictitious IPOs (e.g. 'NextGen AI Solutions') with July–September 2025 dates that have passed. No backend API.

Fix: Use NSE IPO API (/api/ipo/get-upcomingIssue) or third-party feed. Build /api/ipos endpoint in main.py.

Effort: Small–Medium (3–5 days)

● HIGH Sentiment Ensemble S2 Weight — Explicitly Unbuilt — Member 2

What's wrong: ml/sentiment_analyzer.py line 34: self.w2 = 0.00 # Market model (Phase 3 - not implemented yet). Planned third signal zeroed out.

Fix: Define S2 (price momentum, options flow, volume anomaly) and implement, or remove the weight slot.

Effort: Medium (1–2 weeks)

● HIGH No CI/CD Pipeline — Member 1

What's wrong: No .github/workflows/, no Dockerfile for platform backend. Project run manually via start_all.py.

Fix: GitHub Actions: pytest on PR, Docker build, auto-deploy.

Effort: Medium (1–2 days)

● MEDIUM LLM Calibrator — Silent Degradation on Missing Key — Member 2

What's wrong: If GEMINI_API_KEY absent, calibrate() silently returns raw_sentiment. If quota hit (429), calibration permanently disabled for the session. Caller cannot distinguish calibrated vs uncalibrated scores.

Fix: Add calibration_active: bool to sentiment response. Log quota events to monitoring endpoint.

Effort: Small (2–4 hours)

● MEDIUM main.py — 4,989-Line Monolith with Duplicated Logic — Member 1

What's wrong: 56 endpoints, business logic, caching, WS handling, scheduler tasks co-mingled. FII/DII endpoint duplicated in both main.py and News Sentiment backend.

Fix: Extract news proxy, screener, portfolio, market-movers into routers/. Only auth_router.py and trade_router.py extracted so far.

Effort: Medium (1–2 weeks, incremental)

● MEDIUM Celery Beat Schedule Files Committed to Repo — Member 2

What's wrong: celerybeat-schedule (12KB), -shm (32KB), -wal (107KB) are runtime state files committed to git.

Fix: Add to .gitignore, delete from tracking (git rm --cached).

Effort: Small (15 minutes)

● MEDIUM Backend Tests — No Integration or E2E Coverage — Member 1

What's wrong: 10 unit tests using mocks only. Zero integration tests, zero API-level tests, zero E2E tests.

Fix: Add pytest + httpx integration tests for /api/stock-data/chart, /api/screener, /api/trade/place-order. Add conftest.py with test SQLite DB.

Effort: Medium (1–2 weeks)

Cross-Domain Risk Notes

Risk 1: M3 Absent → Screener Appears More Capable Than It Is

The screener returns RSI and volume ratio with presets labelled 'Bullish Breakout' and 'High Momentum'. These are pure threshold filters (`change_pct > 1`, `volume > 100,000`) — not ML signals. Users may assume a recommendation model exists. Once M3 integrates, the distinction will become clear.

Risk 2: Options Integration Will Hit AngelOne WS Subscription Cap

The AngelOne WebSocket subscription cap is ~500 tickers. Options chains require hundreds of strikes per underlying. Before M2 integrates Options, M1 and M2 must coordinate on a token-sharing strategy or a separate data feed — otherwise the subscription pool saturates instantly.

Risk 3: M1 DB Chaos → M3 Cannot Build Signal Pipeline On It

If M3's signal engine tries to read historical OHLCV from the DB, 7+ SQLite files with potentially divergent schemas make it impossible to identify the authoritative source. `stocks_list.csv` (909KB) is also duplicated in two locations and may diverge. Fix the DB before M3 integrates.

Risk 4: Sentiment Scores Not Fed Back Into Live Pipeline

M2's sentiment scores are computed but never incorporated into M1's screener ranking or market movers. There is no shared API contract between M2 and M3 for sentiment-enriched signals. Define this contract now before M3 builds signals that should incorporate sentiment.

Risk 5: JWT Secret Fallback — Silent Auth Compromise

`auth.py` falls back to a hardcoded dev secret if `JWT_SECRET_KEY` is not in `.env`. In any environment where `.env` is not loaded (Docker without env injection, fresh clone), all JWTs issued are forgeable by anyone who reads the source code.

Appendix — API Routes (main.py · 56 Endpoints)

Category	Route	Method	Description
Market Data	/api/stock-data	GET	Daily OHLCV + live candle
	/api/stock-data/chart	GET	Multi-TF chart via ChartService
	/api/stock-data/intraday	GET	Intraday candles with backfill
	/api/stock-data/range	GET	Date-range OHLCV query
	/api/stock-data/weekly	GET	Weekly resampled candles
	/api/stock-data/monthly	GET	Monthly resampled candles
Live Prices	/api/live-prices	POST	Batch live price fetch for ticker list
	/api/market-movers	GET	Top gainers, losers, most active
	/api/sector-leaders	GET	Top stock per sector
	/api/screener	GET	Full screener with RSI, vol ratio, filters
Trading	/api/trade/place-order	POST	Open position (CSRF protected)
	/api/trade/close-position	POST	Close position manually
	/api/trade/set-limits	POST	Update TP/SL on open position
Portfolio	/api/portfolio/open-positions	GET	List open positions
	/api/portfolio/closed-positions	GET	Closed position history
	/api/portfolio/summary	GET	Balance, P&L, metrics
	/api/portfolio/transactions	GET	Transaction log
News	/api/news/general	GET	General market news (Google RSS)
	/api/news/ticker/{ticker}	GET	Ticker-specific news
	/api/scanx/news/full/all	GET	Proxy → News Sentiment :8003
	/api/scanx/news/full/{ticker}	GET	Proxy → News Sentiment :8003
	/api/scanx/news/market-sentiment	GET	Aggregate market sentiment score
Auth	via auth_router.py	POST	register, login, logout, verify-email, reset-password
WebSockets	/ws/dashboard	WS	Live tick broadcast to all clients
	/ws/user	WS	Private TP/SL fill push per user
Health	/api/health	GET	System health, WS status, yfinance stats
	/api/monitoring/candle-system	GET	Aggregator, cache, recovery health
	/api/validation/run	GET	Run OHLC + volume validation for ticker
Utility	/api/all-stocks	GET	Full stock metadata list (1h cache)
	/api/watchlist	GET	User watchlist
	/api/fii-dii	GET	FII/DII data (fragile scraping)
	/api/crypto-prices	GET	Top crypto via CoinGecko/yfinance

Category	Route	Method	Description
	/api/time	GET	Server UTC epoch for TradingView

Appendix — Critical Environment Variables

Variable	Required By	Risk If Missing
ANGELONE_API_KEY	angelone_service.py	Live data unavailable; falls back to yfinance only
ANGELONE_CLIENT_ID	angelone_service.py	Same as above
ANGELONE_PASSWORD	angelone_service.py	Same as above
ANGELONE_TOTP_TOKEN	angelone_service.py	Same as above
JWT_SECRET_KEY	auth.py	Falls back to hardcoded dev key — all JWTs forgeable
DATABASE_URL	database.py	Falls back to SQLite — unclear which file is authoritative
GEMINI_API_KEY	ml/llm_calibrator.py	LLM calibration silently disabled
REDIS_URL	tasks/celery_app.py	Celery workers fail; no background news scraping
SENTRY_DSN	main.py	Error tracking disabled (non-critical)

Appendix — Stub / TODO Markers Found

PowerShell grep search on backend/*.py: Only SQL parameter placeholder strings found. Zero TODO, FIXME, NotImplementedError, or pass-as-stub markers in any backend/*.py file. The platform backend is fully implemented.

Explicit stub found in ML codebase:

```
# ml/sentiment_analyzer.py line 34
self.w2 = 0.00 # Market model (Phase 3 - not implemented yet)
```

Appendix — Database Tables (models.py)

Table Name	Purpose
stock_data	Daily historical OHLCV data
intraday_ticks	Raw price ticks (legacy)
current_day_candle	Live intraday summary of the current day
intraday_candles_1min/5min/15m/30m/1h	Legacy per-timeframe candles table
candles	Unified timeframe table (new design — both coexist)
market_sessions	Session type and session time definitions
holidays	NSE holidays calendar
retention_jobs	Log of rolling data retention conversion cleanups
stock_metadata	Company description, logo, sector mapping, premium flag
users / login_attempts / verification_tokens	Authentication and brute-force lock system tables
positions / orders / transactions	Paper trading simulation engine data
user_watchlist / user_chart_settings	Personalization preferences settings

Appendix — File Count Summary by Area

Development Domain	Python Files	HTML/JS Files	Test Files
Platform Backend	~183	—	10 (unit tests with mocks)
Platform Frontend	—	~76	0
News Sentiment Backend	~70	—	0 (20+ debug scripts)
News Sentiment ML	9	—	0
News Sentiment Frontend	—	3	0
Options Intelligence	0 (Missing)	0 (Missing)	0
Quant Research	0 (Missing)	0 (Missing)	0